

An aerial night view of a city, likely Shanghai, with a network of glowing white lines and nodes overlaid on the image, symbolizing wireless networks. The city lights are visible against a dark blue sky.

Wireless Networks

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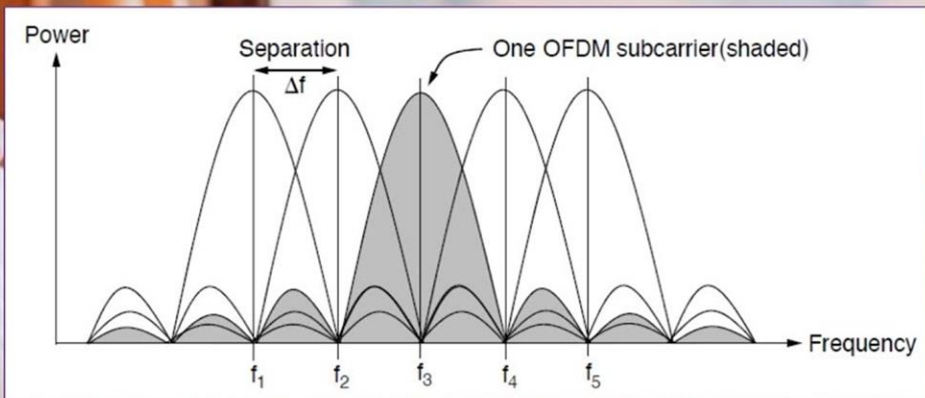
Diversity Techniques

Lecture 7



Contents

- ✓ Multiple Input Multiple Output (MIMO)
- ✓ Orthogonal Frequency Division Multiplexing (OFDM)



OFDM

Orthogonal Frequency Division Multiplexing



Diversity in Wireless Communication

Recap of Wireless Communication Channel Characteristics

■ Channel Characteristics & Performance

- Wireless system performance depends on the transmitter-receiver channel conditions.
- Key terminologies: **coherence bandwidth** and **coherence time**.

■ Coherence Bandwidth (B_c)

- Frequency range where the channel response is flat.
- Inversely proportional to the delay spread of the channel.
- If signal bandwidth $B_s \leq B_c$ → No distortion (flat fading).
- If signal bandwidth $B_s > B_c$ → Distorted signal (selective fading).

■ Coherence Time (T_c)

- Approximate time duration for which the channel remains constant.
- Inversely proportional to the Doppler frequency shift (B_d).
- If coherence time (T_c) $\geq$ symbol time → Slow fading.
- If coherence time (T_c) $<$ symbol time → Fast fading.

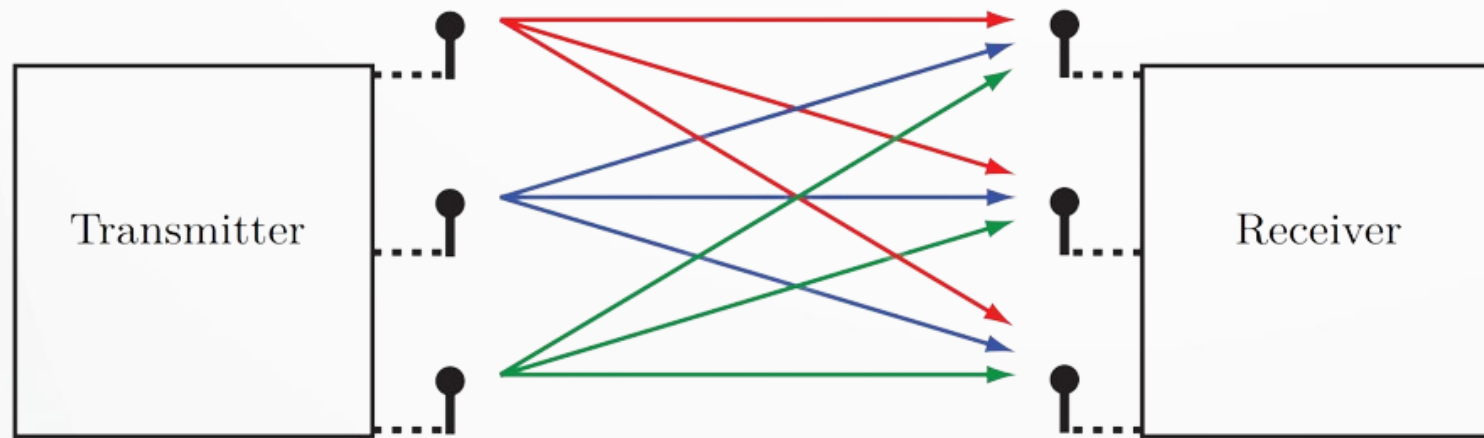
Recap- Diversity in Wireless Communication

Diversity

- Diversity is a technique used in wireless communication to:
 - improve signal **reliability**
 - reduce the effects of **fading**, **interference**, and **signal degradation**.
- It involves transmitting or receiving the same information over **multiple channels** that **experience independent fading**.
- This increases the probability of receiving the signal **correctly**, even if some paths are affected by fading or noise.

■ Multiple Input Multiple Output (MIMO)

- ☑ MIMO system uses multiple transmit antennas and multiple receive antennas to improve **data rate**, **reliability** and **spectral** efficiency



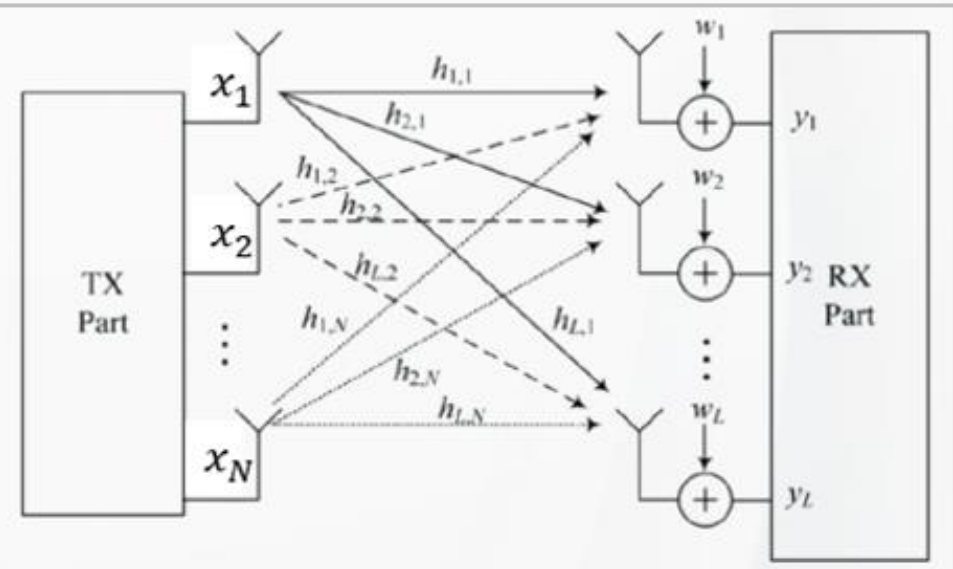
- ☑ MIMO is the foundation of modern wireless standards, including:
 - ✓ 4G LTE, LTE advanced, 5G and 6G
 - ✓ Wi-Fi (802.11n/ac/ax)
 - ✓ Massive MIMO systems (**out of scope**)

■ Multiple Input Multiple Output (MIMO)

☑ Mathematical modelling of MIMO

- It consists of N transmit antennas and L receive antennas.
- Each channel response is represented by $h_{m,k}$, where:
 - k denotes the transmit antenna ($k = 1, \dots, N$).
 - m denotes the receive antenna ($m = 1, \dots, L$).
- The channel response $h_{m,k}$ describes the transmission from antenna k to antenna m .
- The received signal at a given antenna m (for $m = 1, 2, 3, \dots, L$) becomes

$$y_m = \sum_{k=1}^N h_{m,k} x_k + n_m$$



■ Multiple Input Multiple Output (MIMO)

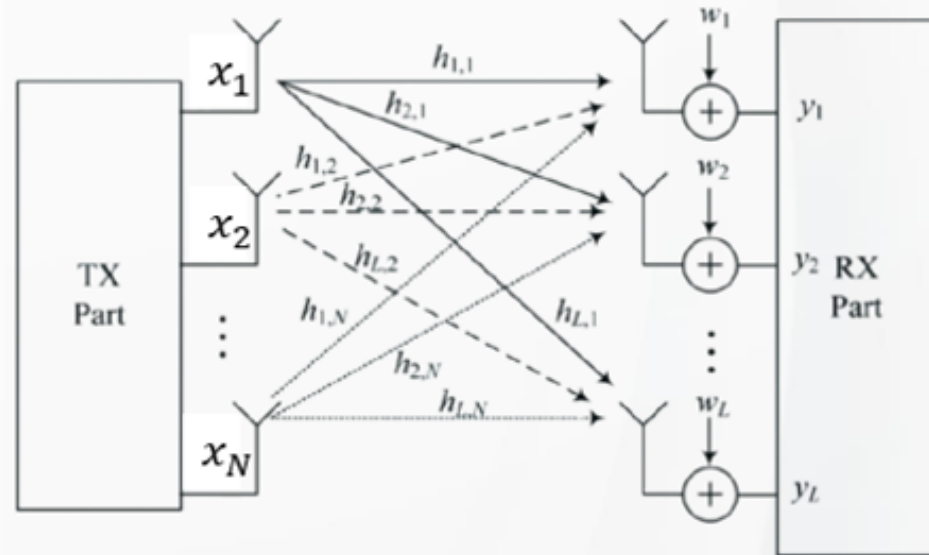
☑ Mathematical modelling of MIMO

- This system model can be written in a concise matrix form as

$$\begin{bmatrix} y_1 \\ \vdots \\ y_L \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^N h_{1,k} x_k \\ \vdots \\ \sum_{k=1}^N h_{L,k} x_k \end{bmatrix} + \begin{bmatrix} n_1 \\ \vdots \\ n_L \end{bmatrix}$$
$$= \begin{bmatrix} h_{11} & \dots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{L1} & \dots & h_{LN} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} n_1 \\ \vdots \\ n_L \end{bmatrix}$$

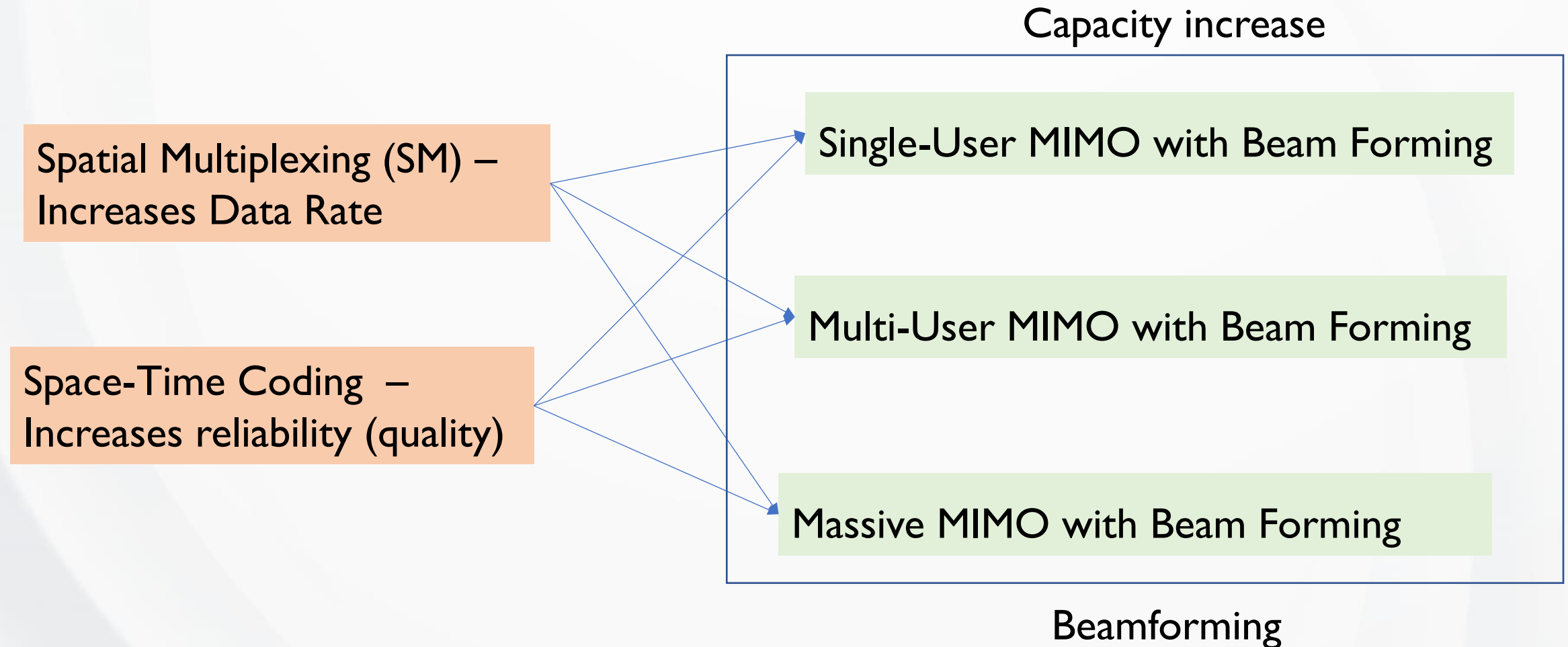
$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_L \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} \quad \mathbf{n} = \begin{bmatrix} n_1 \\ \vdots \\ n_L \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} h_{11} & \dots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{L1} & \dots & h_{LN} \end{bmatrix}$$



■ Multiple Input Multiple Output (MIMO)

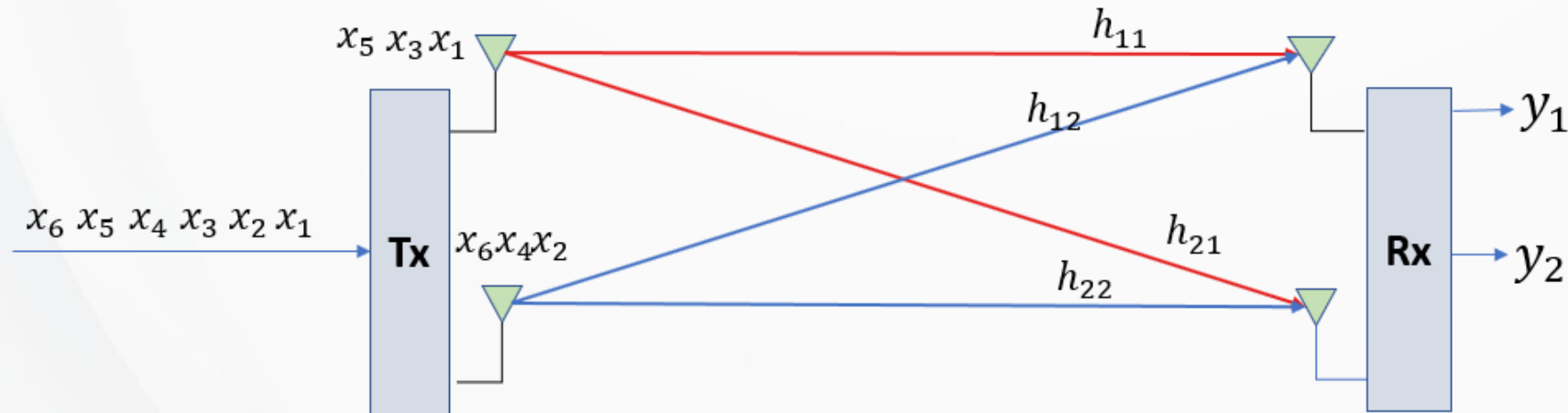
☑ MIMO systems



■ Multiple Input Multiple Output (MIMO)

☑ Spatial Multiplexing (SM) in MIMO Systems

- **Spatial Multiplexing (SM)** is a MIMO technique that increases data rates by transmitting multiple independent data streams simultaneously over multiple spatial channels. It takes advantage of multiple antennas at both the transmitter and receiver to send separate data streams, improving spectral efficiency.



- If the data rate is R then the Tx will send with rate $2R$.
- If the number of Tx antennas is N then the data rate will be NxR

■ Multiple Input Multiple Output (MIMO)

☑ Limitations on the Number of Antennas in MIMO Systems

While increasing the number of antennas in a MIMO system can enhance performance, several practical and theoretical limitations constrain their scalability:

❖ Channel Correlation

- **Issue:** When antennas are placed too close to each other, the channels experience **high spatial correlation**, reducing MIMO benefits.
- **Impact:** The system loses diversity and multiplexing gains, limiting capacity improvements.
- **Solution:** Antennas must be spaced at least half the wavelength ($\lambda/2$) apart to maintain independent channels. **However**, spacing requirement limits the number of antennas in small-sized devices.

■ Multiple Input Multiple Output (MIMO)

☑ Limitations on the Number of Antennas in MIMO Systems

❖ Hardware Complexity & Cost

- **Issue:** More antennas require additional RF chains, increasing cost, power consumption, and system complexity.
- **Impact:** High deployment and operational expenses, especially in mobile and IoT applications.
- **Solution:** Advanced antenna design and hybrid beamforming can reduce hardware demands.

❖ Power and Energy Consumption

- **Issue:** Each antenna consumes power for transmission, reception, and signal processing.
- **Impact:** Large MIMO systems may exceed feasible power budgets, especially in battery-powered devices.
- **Solution:** Energy-efficient precoding and power allocation techniques.

■ Multiple Input Multiple Output (MIMO)

☑ Limitations on the Number of Antennas in MIMO Systems

❖ Signal Processing & Computational Complexity

- **Issue:** Larger MIMO systems require high-dimensional matrix computations for detection and precoding at Rx.

In a MIMO system, the received signal $\mathbf{y}$ is modeled as:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

Where:

- $\mathbf{y}$ = Received signal vector ($M \times 1$)
- $\mathbf{H}$ = Channel matrix ($M \times K$)
- $\mathbf{x}$ = Transmitted signal vector ($K \times 1$)
- $\mathbf{n}$ = Noise vector ($M \times 1$)

To recover the transmitted signal $\mathbf{x}$, we need to invert the effect of the channel matrix $\mathbf{H}$.

■ Multiple Input Multiple Output (MIMO)

☑ Limitations on the Number of Antennas in MIMO Systems

❖ Signal Processing & Computational Complexity- Open loop MIMO system

- **Computation at Receiver:** the receiver estimates channel response but does not provide feedback to the transmitter.

Approach:

- If $\mathbf{H}$ is a square matrix ($M = K$) and invertible, we can compute its inverse:

$$\hat{\mathbf{x}} = \mathbf{H}^{-1} \mathbf{y}$$

which gives an estimate $\hat{\mathbf{x}}$ of the transmitted signal.

- If $\mathbf{H}$ is not square ($M \neq K$), we use the pseudo-inverse:

$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

where $\mathbf{H}^H$ is the Hermitian (conjugate transpose) of $\mathbf{H}$.

**Zero-Forcing (ZF) Detection
(Inverse of $\mathbf{H}$)**

- **Impact:** Increases processing time and power consumption as the channel response matrix increases.
- **Solution:** Efficient detection algorithms (e.g., MMSE, sphere decoding) and massive MIMO approximations.

■ Multiple Input Multiple Output (MIMO)

☑ Limitations on the Number of Antennas in MIMO Systems

❖ Signal Processing & Computational Complexity - Closed loop MIMO system

- **Computation at Transmitter:** the receiver estimates the channel response and provides feedback to the transmitter. As a result, Channel State Information (CSI) is available at both the transmitter and receiver.

☑ At the transmitter:

- 1) Channel estimation (from feedback)
- 2) Precoding / Beamforming based on CSI
- 3) Adaptive modulation and coding based on channel quality

☑ At the receiver:

- 1) Channel estimation (for decoding)
- 2) MIMO detection (like ZF, MMSE, Sphere Decoding)
- 3) Error correction decoding (e.g., Turbo Codes, LDPC)

■ Multiple Input Multiple Output (MIMO)

☑ Limitations on the Number of Antennas in MIMO Systems

When to Use Closed-Loop vs. Open-Loop MIMO?

Feature	Closed-Loop MIMO	Open-Loop MIMO
CSI at Transmitter	Yes (via feedback)	No
Channel Estimation	At receiver & fed back to transmitter	Only at receiver
Feedback Required?	Yes (adds overhead)	No
Performance	Better (higher capacity, optimized beamforming)	Lower performance
Use Case	Low mobility (slow fading channels, e.g., indoor WiFi, 5G base stations)	High mobility (fast fading, e.g., mobile users, vehicular networks)
Techniques Used	Precoding (e.g., Eigen beamforming, ZF, MMSE)	Spatial diversity (e.g., STBC, Alamouti)

■ Multiple Input Multiple Output (MIMO)

✓ Closed loop MIMO system numerical example

Let's consider a **2×2 MIMO system** with **two transmit antennas** and **two receive antennas**. We will compute **Zero-Forcing (ZF) Precoding** as an example.

1. Define the Channel Matrix H

Let's assume the channel matrix H is: $H = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

2. Compute the ZF Precoding Matrix W

Zero-Forcing precoding uses: $W = (H^H H)^{-1} H^H$

Step 1: Compute H^H (Hermitian/Transpose of H)

$$H^H = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Step 2: Compute $H^H H$

$$H^H H = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+9 & 2+12 \\ 2+12 & 4+16 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$

Step 3: Compute $(H^H H)^{-1}$ (Inverse of $H^H H$)

The inverse of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is: $A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Applying this to our matrix:

$$\begin{aligned} (H^H H)^{-1} &= \frac{1}{(10 \times 20 - 14 \times 14)} \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix} \\ &= \frac{1}{200 - 196} \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -3.5 \\ -3.5 & 2.5 \end{bmatrix} \end{aligned}$$

Step 4: Compute $W = (H^H H)^{-1} H^H$

$$W = \begin{bmatrix} 5 & -3.5 \\ -3.5 & 2.5 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} (5 \times 1 + -3.5 \times 2) & (5 \times 3 + -3.5 \times 4) \\ (-3.5 \times 1 + 2.5 \times 2) & (-3.5 \times 3 + 2.5 \times 4) \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

■ Multiple Input Multiple Output (MIMO)

☑ Closed loop MIMO system numerical example

Let's consider a **2×2 MIMO system** with **two transmit antennas** and **two receive antennas**. We will compute **Zero-Forcing (ZF) Precoding** as an example.

3. Transmit Precoded Signal

If the original transmitted signal is:

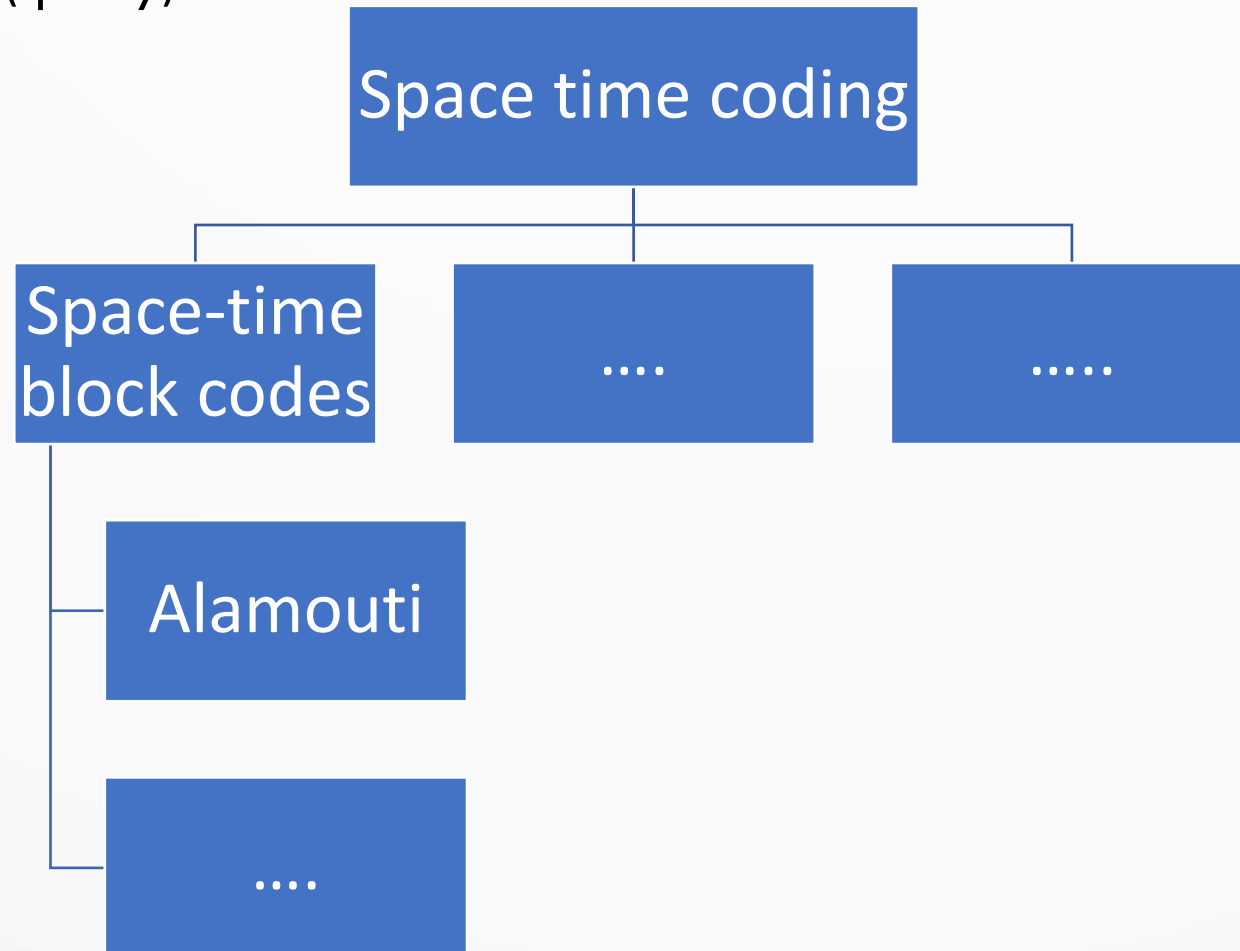
$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} x' = Wx &= \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} (-2 \times 1 + 1 \times 1) \\ (1.5 \times 1 - 0.5 \times 1) \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 1 \end{bmatrix} \end{aligned}$$

■ Multiple Input Multiple Output (MIMO)

☑ Space-Time Coding – Increases quality

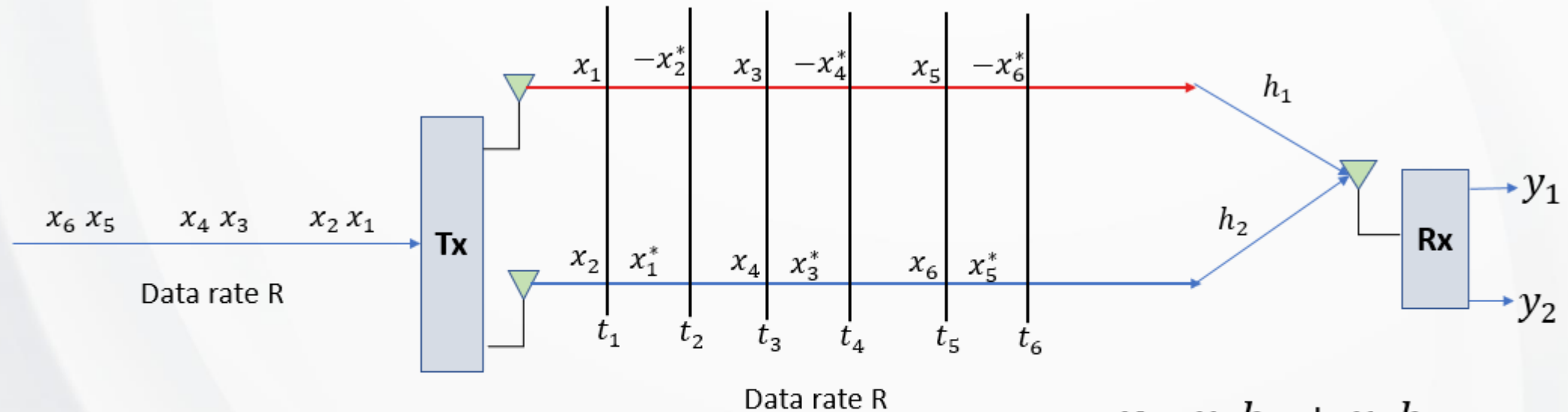
- **Space-Time Coding (STC)** is a technique used in Multiple Input Multiple Output (MIMO) systems to enhance reliability (quality).



■ Multiple Input Multiple Output (MIMO)

☑ Space-Time Coding – Increases quality

- **Space-Time Coding (STC)** is a technique used in Multiple Input Multiple Output (MIMO) systems to enhance reliability (quality).



At t_1 $y_1 = x_1 h_1 + x_2 h_2$

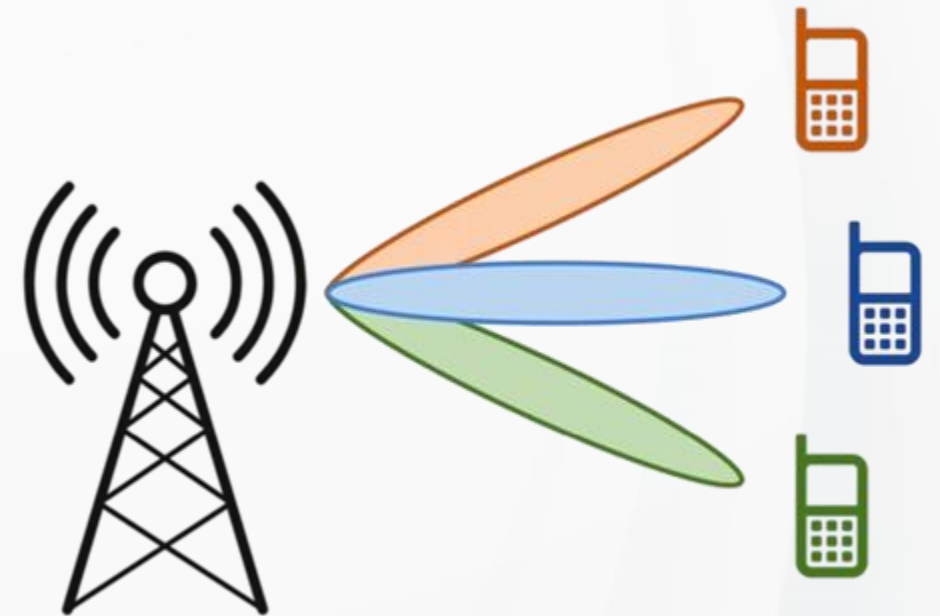
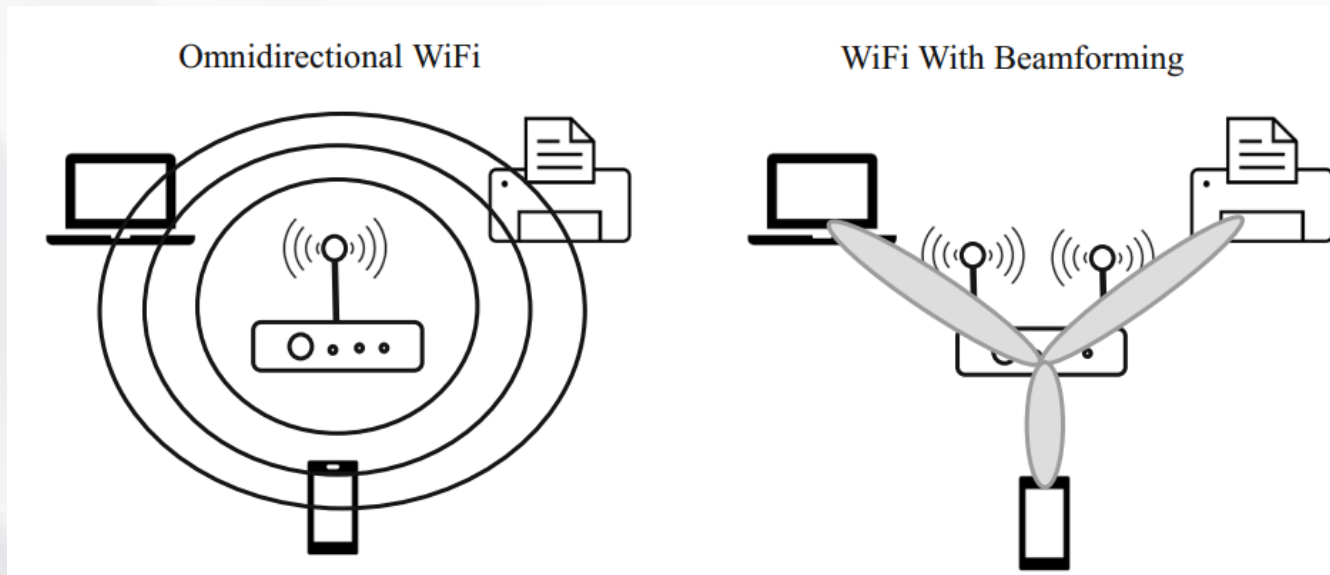
At t_2 $y_2 = -x_2^* h_1 + x_1^* h_2$

Then use Alamouti algorithm to extract x_1 and x_2

Multiple Input Multiple Output (MIMO)

Beamforming

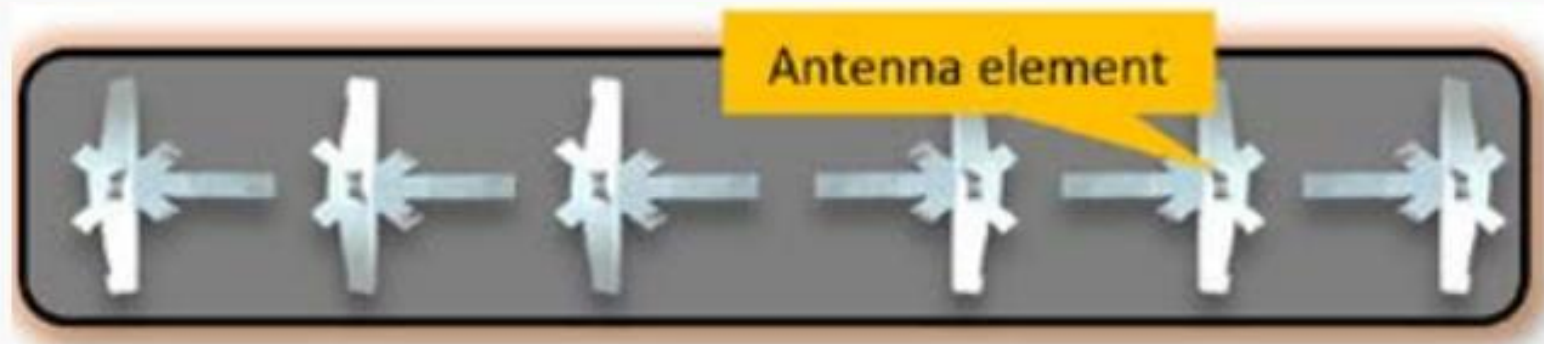
Beamforming is a signal processing technique used in wireless communication systems to direct the transmitted or received signal toward a specific direction, rather than broadcasting it in all directions. It is widely used in MIMO (Multiple Input Multiple Output) systems, massive MIMO, and 5G networks to enhance signal strength, reduce interference, and improve spectral efficiency.



Multiple Input Multiple Output (MIMO)

What is Phased Array Beamforming?

Phased array beamforming is a signal processing technique where multiple antennas (an antenna array) are used to steer a signal beam without physically moving the antennas. This is achieved by applying controlled phase shifts to the signals transmitted or received by each antenna element.



Orthogonal Frequency Division Multiplexing (OFDM)

Orthogonal Frequency Division Multiplexing (OFDM)

☑ A **narrowband channel** is one where signal bandwidth $B_s \leq B_c$
→ No distortion (flat fading).

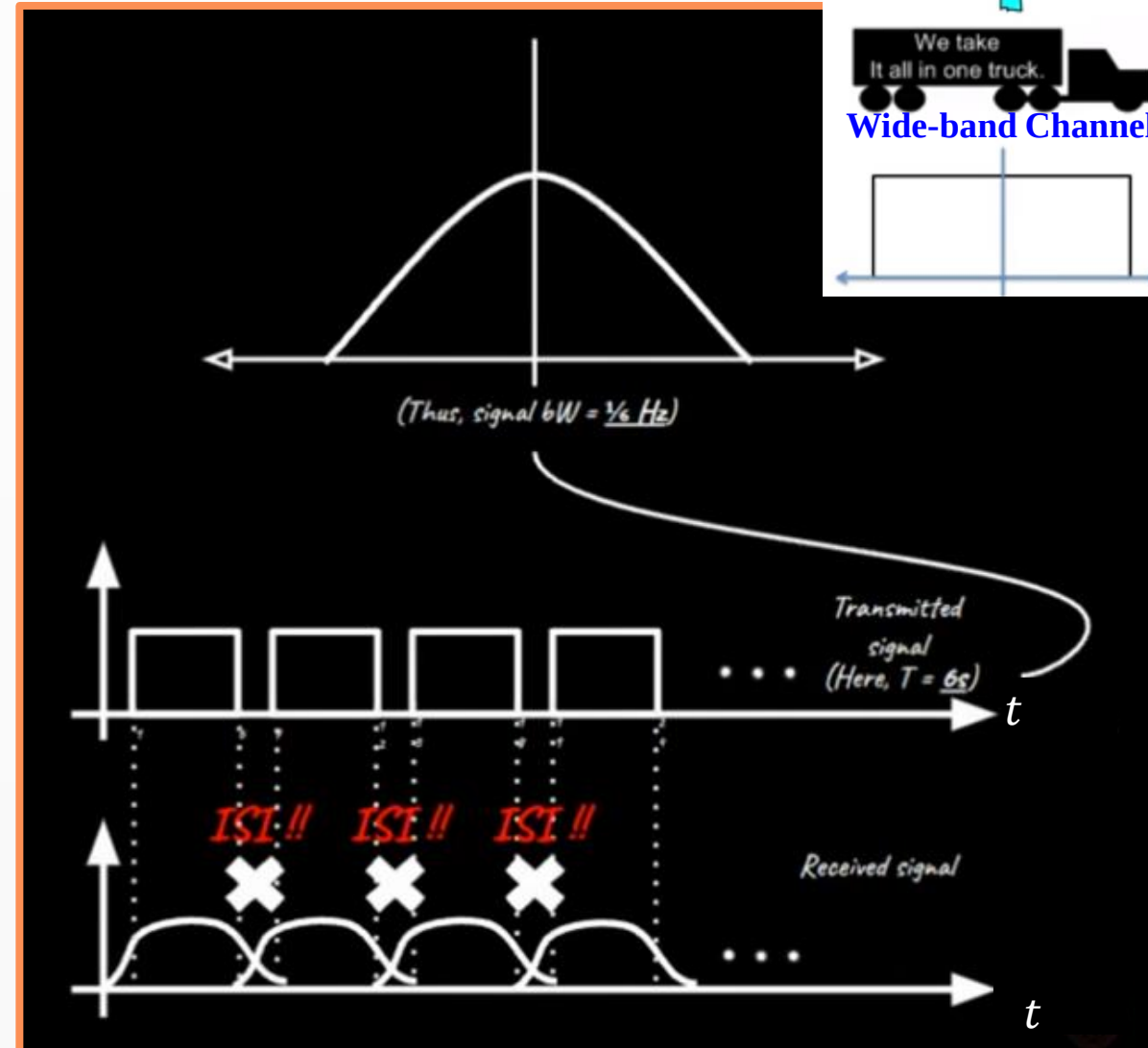
☑ A **wideband channel** is one where signal bandwidth $B_s > B_c$
which leads to:

- 1) **frequency-selective fading**, meaning different frequency components of the signal experience different gains.
- 2) The channel cannot be represented by a single constant h but instead has a frequency-dependent response $H(f)$.
- 3) **Due to multipath effects**, different copies of the signal arrive at different delays, causing **inter-symbol interference (ISI)**.

Orthogonal Frequency Division Multiplexing (OFDM)

☑ Single carrier transmission

- ❖ In single carrier transmission, data is modulated onto a single frequency carrier.
- ❖ Due to multipath (reflection from buildings, trees, and other obstacles), multiple copies of the signal that arrive at different times.
- ❖ This leads to spreading of signals in time domain.
- ❖ If the transmitted symbols are not spaced far enough, this spreading leads to **ISI** problem
- ❖ This makes it difficult for the receiver to distinguish between individual symbols, leading to errors



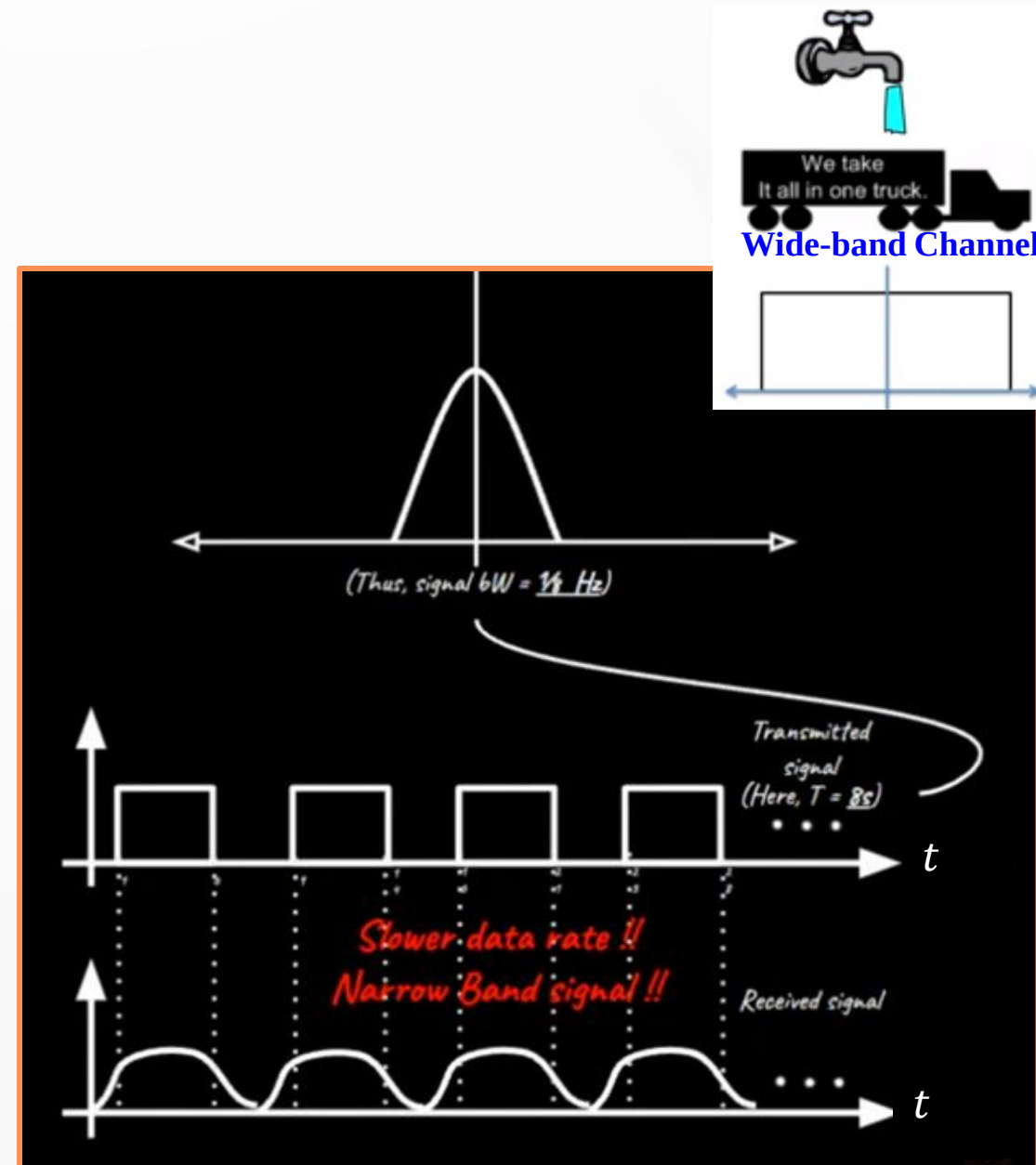
Orthogonal Frequency Division Multiplexing (OFDM)

☑ Single carrier transmission

- ❖ An early and practical solution to ISI was the introduction of a guard space to separate symbols and prevent overlap due to multipath effects.
- ❖ This has the same effects as:
 - Increasing the symbol time
 - Decreasing the signal bandwidth

✗ Disadvantages:

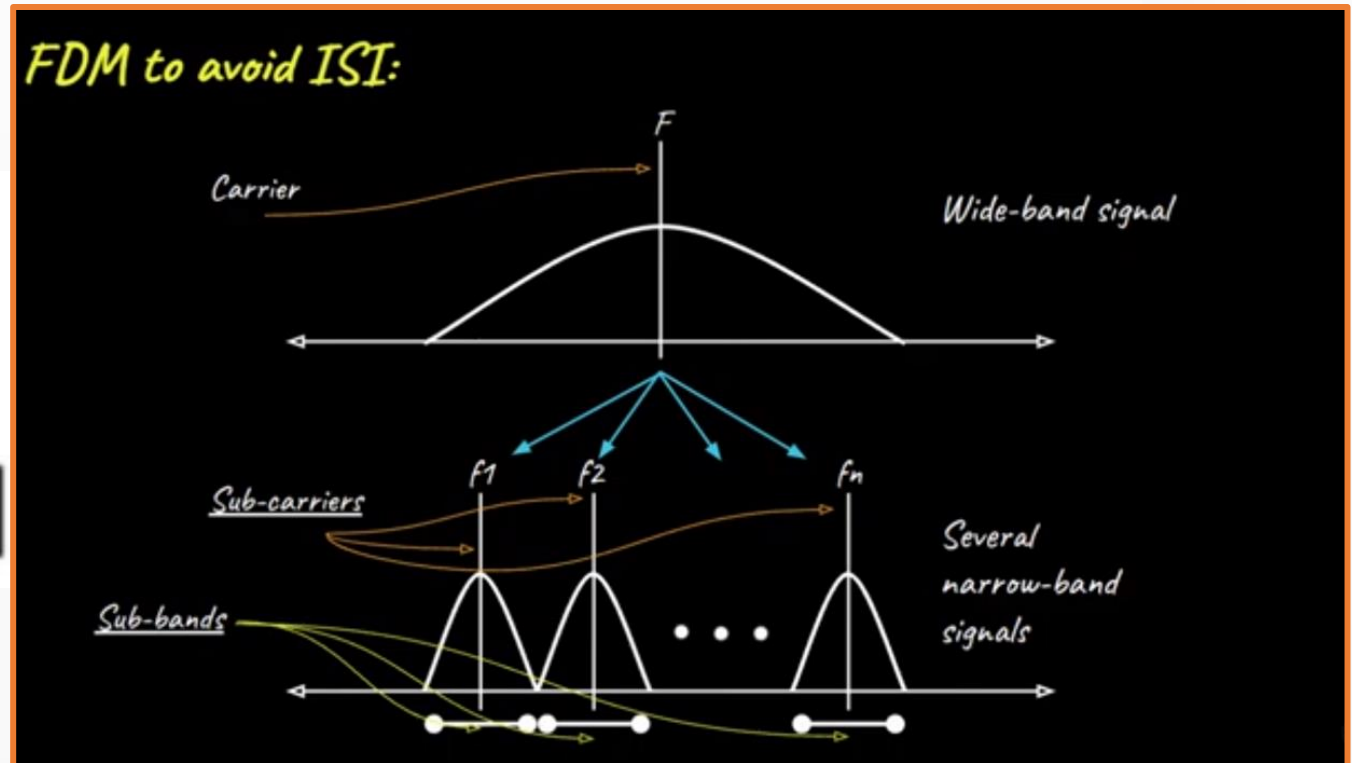
- Reduces spectral efficiency (since guard time does not carry useful data).
- Does not eliminate ISI completely (if the delay spread is longer than the guard interval).



Orthogonal Frequency Division Multiplexing (OFDM)

☑ Multi-carrier transmission

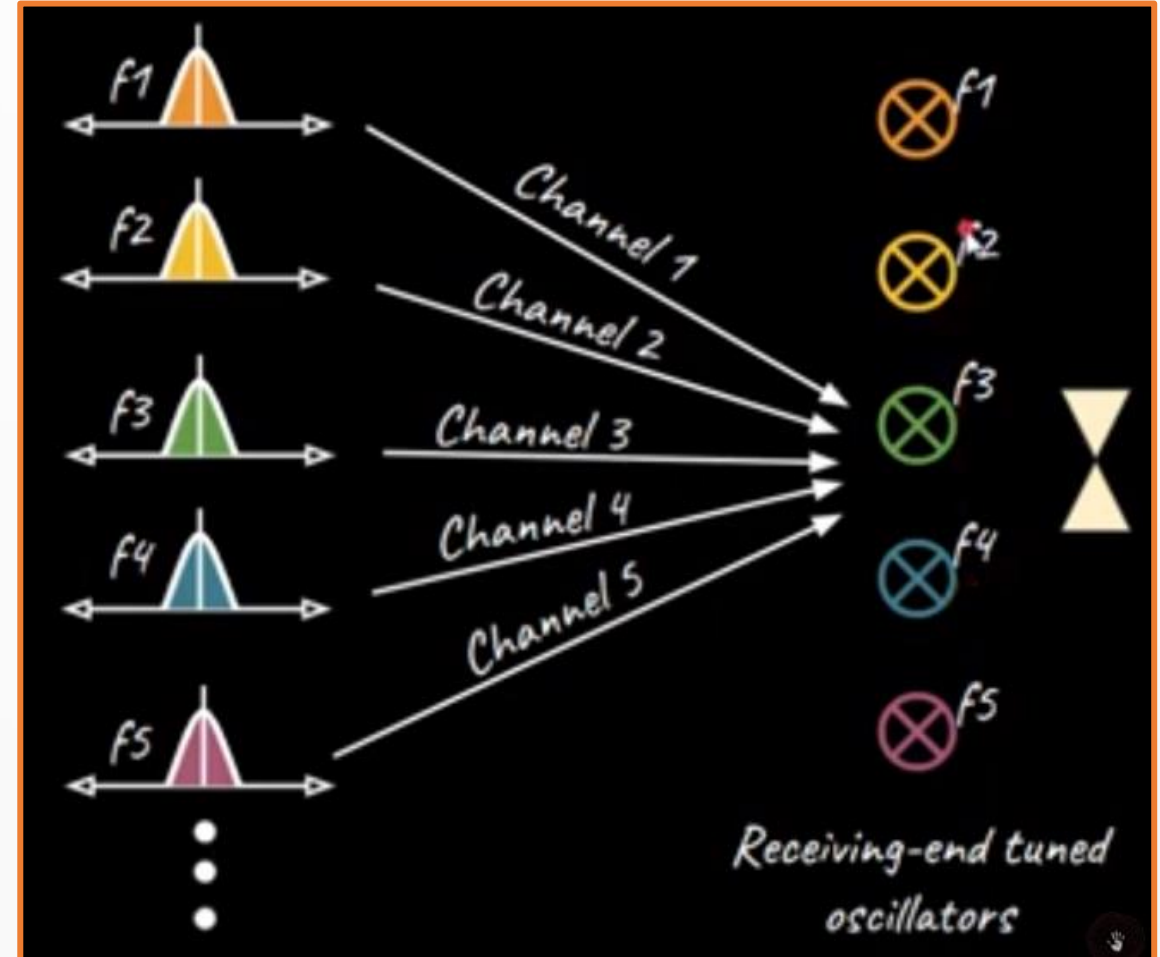
- The goal now is to eliminate ISI while maintaining the data rate.
- Frequency Division Multiplexing (FDM) is a technique where multiple signals are transmitted simultaneously over different frequency bands.



Orthogonal Frequency Division Multiplexing (OFDM)

✓ Multi-carrier transmission

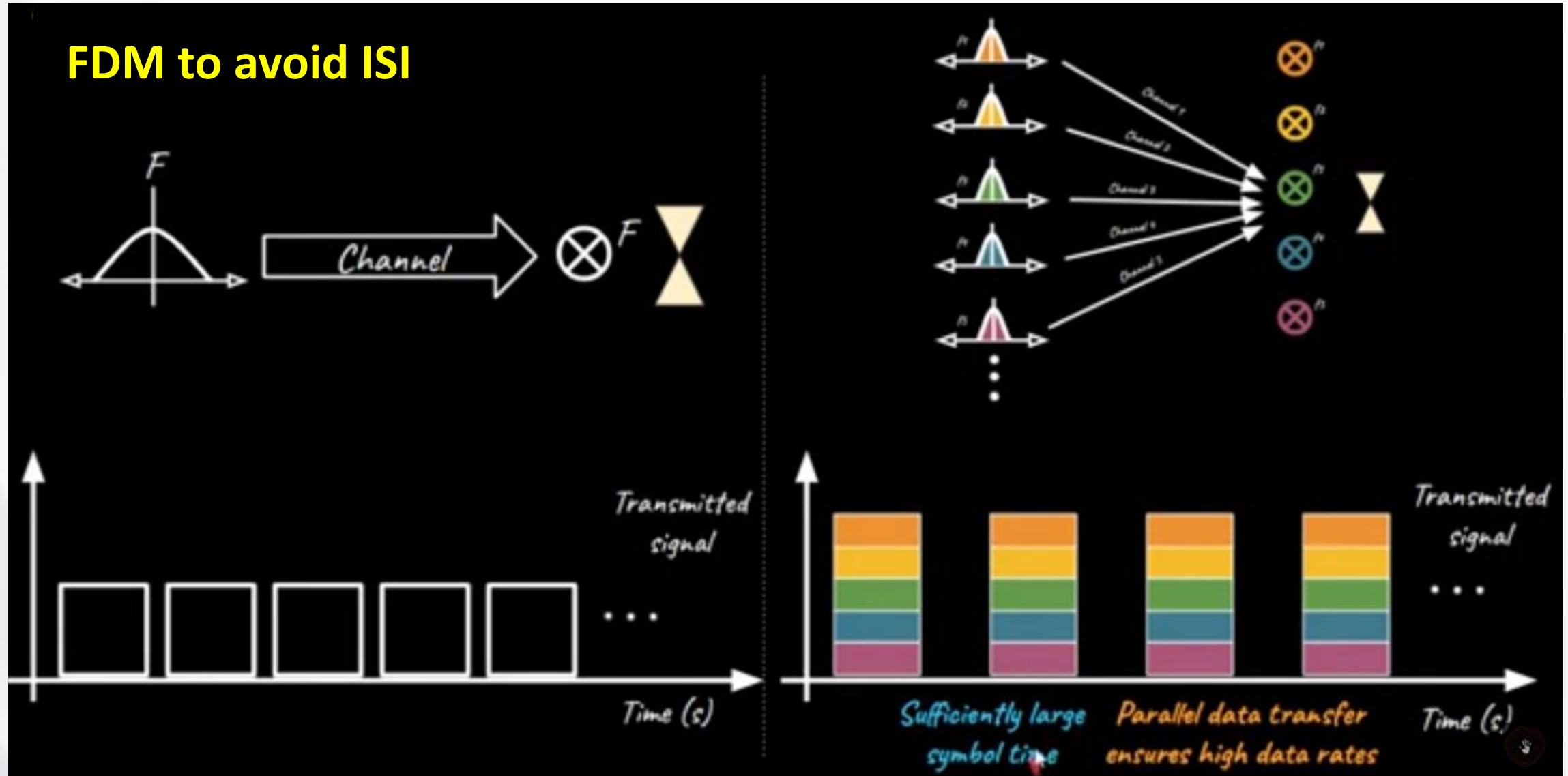
- These modulated signals use different carrier known as sub-carriers and merged in a linear summing circuit to form a composite signal for transmission.
- At the receiver, the signals are surpassed by band-pass filters to isolate the individual frequency channels.



Orthogonal Frequency Division Multiplexing (OFDM)

☑ Multi-carrier transmission

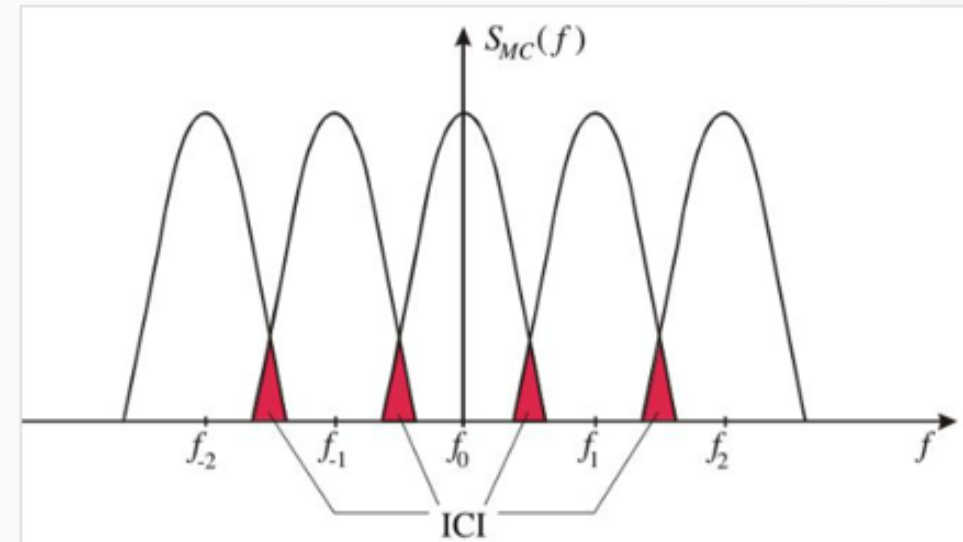
FDM to avoid ISI



Orthogonal Frequency Division Multiplexing (OFDM)

✓ Multi-carrier transmission

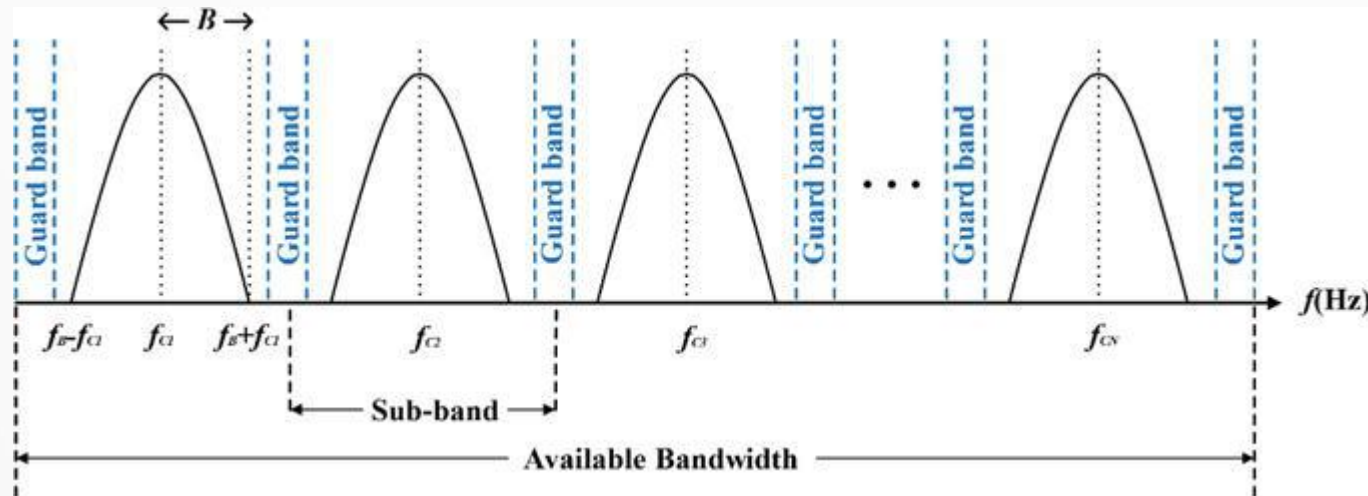
- Ideally, these carriers are spaced sufficiently apart to prevent overlap.
- However, in practical scenarios, factors such as Doppler shifts (caused by relative movement between transmitter and receiver) or oscillator phase noise can cause the carriers to spread beyond their allocated bands, resulting in overlap and interference.
- This overlap leads to Inter-Carrier Interference (ICI).
- ICI **degrades** the signal quality and makes it harder for the receiver to distinguish between the **data on different sub-carriers**. This leads to increased **error rates** and **reduced system performance**.



Orthogonal Frequency Division Multiplexing (OFDM)

✓ Multi-carrier transmission

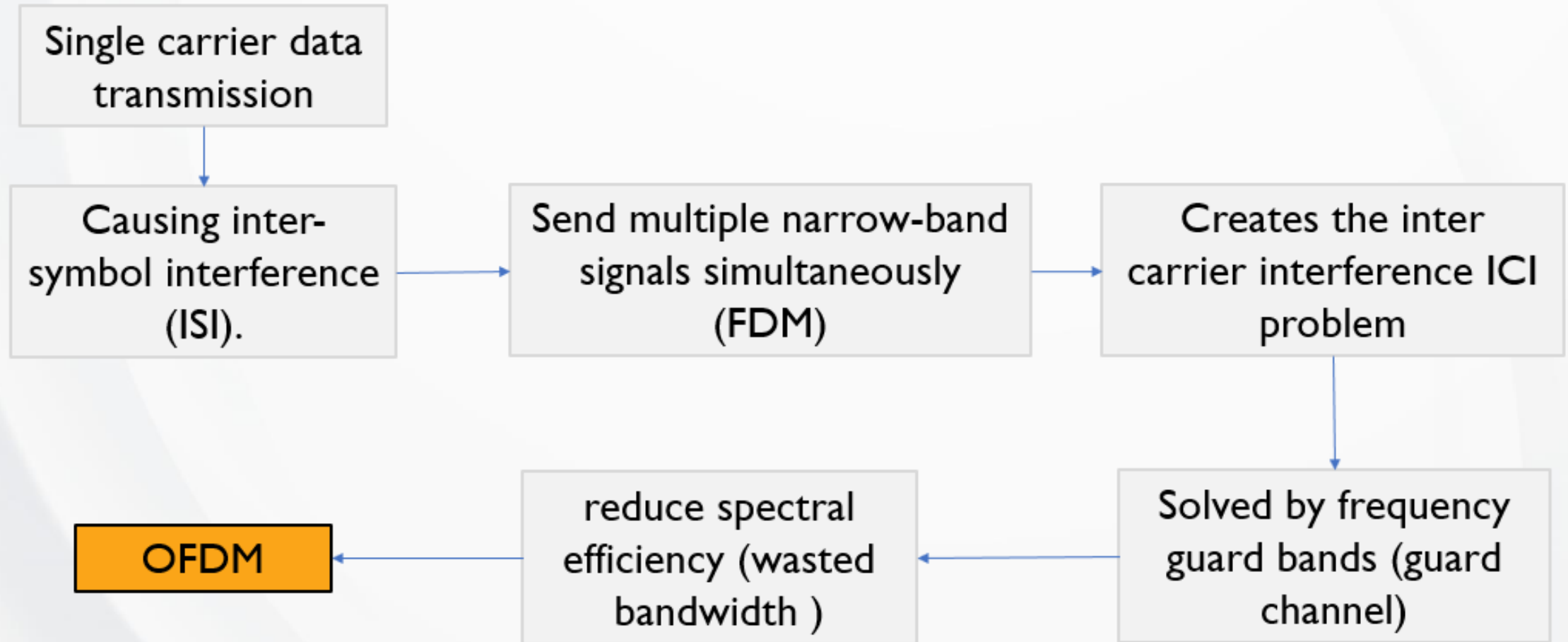
- **Frequency Guard Bands:** Increasing the size of the guard bands between the frequency allocations of different users can help to reduce ICI.



- Guard bands reduce spectral efficiency.
- They are unused frequency portions placed between users to prevent interference.
- This results in wasted bandwidth that could otherwise be used for data transmission.

Orthogonal Frequency Division Multiplexing (OFDM)

✓ Moving towards OFDM



Orthogonal Frequency Division Multiplexing (OFDM)

☑ OFDM, or **O**rthogonal **F**requency **D**ivision **M**ultiplexing, is a modulation technique used in modern communication systems. To fully grasp the concept of OFDM, it is essential to understand the meaning of its key components:.

1) Orthogonal:

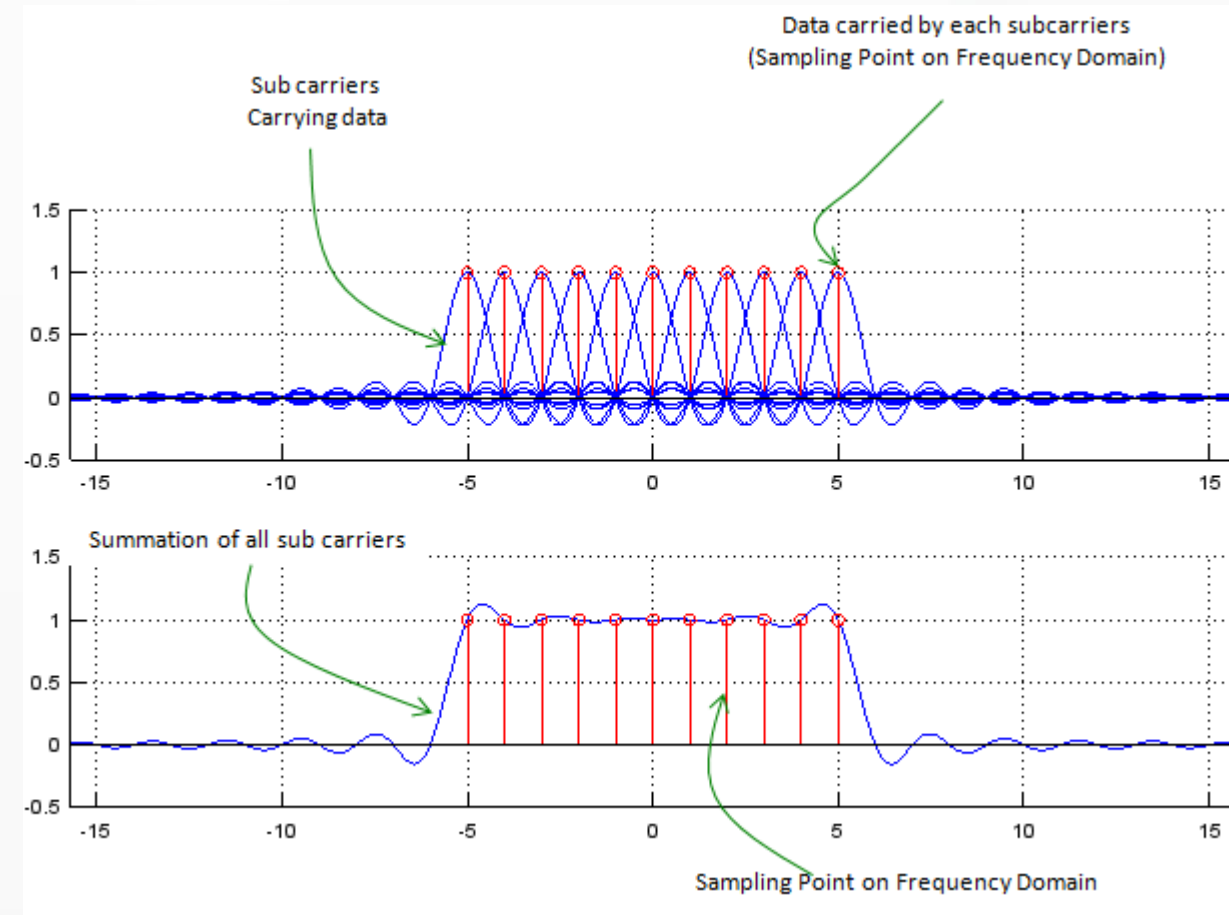
- Two signals are orthogonal if their inner product (integral over time) is zero.
- Mathematically: For two sinusoids with different frequencies f_1 and f_2 , their inner product over a time interval T is zero if they are orthogonal.

2) **Frequency Division**: Dividing the available bandwidth into separate frequency bands, with each signal transmitted on its own band simultaneously.

3) **Multiplexing**: Combining multiple signals to share a single communication channel, allowing each signal to be separated and recovered at the receiver.

Orthogonal Frequency Division Multiplexing (OFDM)

- ☑ OFDM is a technology that divides a wide frequency band into multiple smaller frequencies, called subcarriers.
- ☑ Data is transmitted on each subcarrier separately.
 - This division of a wide frequency band into narrow subcarriers represents **Frequency Division**.
 - Since all subcarriers transmit data simultaneously, this process is a form of **Multiplexing**.



Orthogonal Frequency Division Multiplexing (OFDM)

Subcarrier Division in OFDM

☑ How small can each subcarrier be?

- Example: A 1 MHz bandwidth can be split into subcarriers.
- 1000 subcarriers (1 kHz interval): Transmits 1000 bits simultaneously.
- 100 subcarriers (10 kHz interval): Transmits 100 bits at a time.

☑ Why not use too many subcarriers?

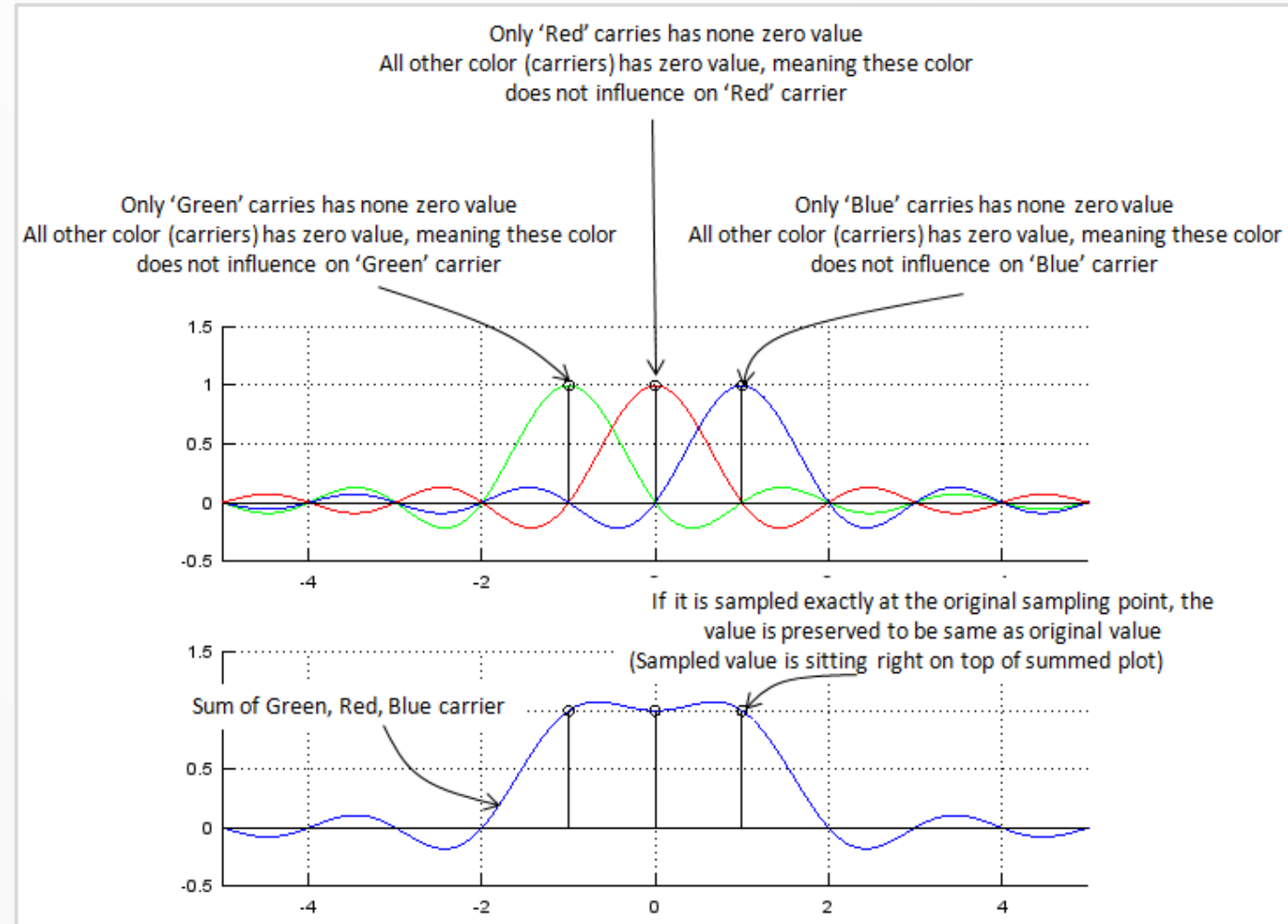
- More subcarriers (smaller intervals): Increases data rate but causes interference.
- Fewer subcarriers (larger intervals): Reduces interference but decreases data rate.

Orthogonal Frequency Division Multiplexing (OFDM)

Subcarrier Division in OFDM

☑ OFDM's Solution: Orthogonality

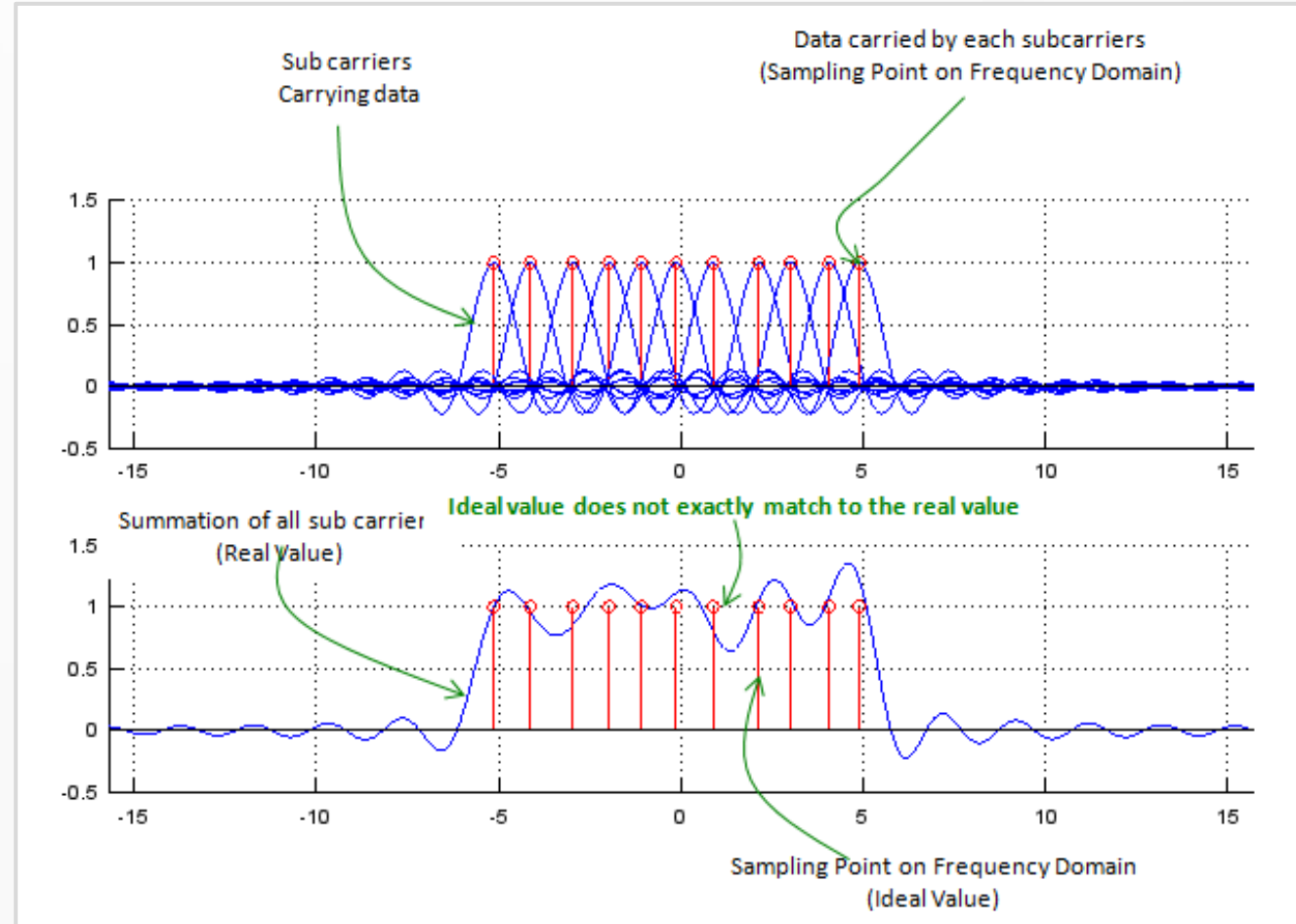
- Subcarriers are arranged so that at each sampling point, only one has a non-zero value.
- This ensures no interference between subcarriers.
- "Orthogonal" means "Independent" → No Interaction/Interference.



Orthogonal Frequency Division Multiplexing (OFDM)

OFDM

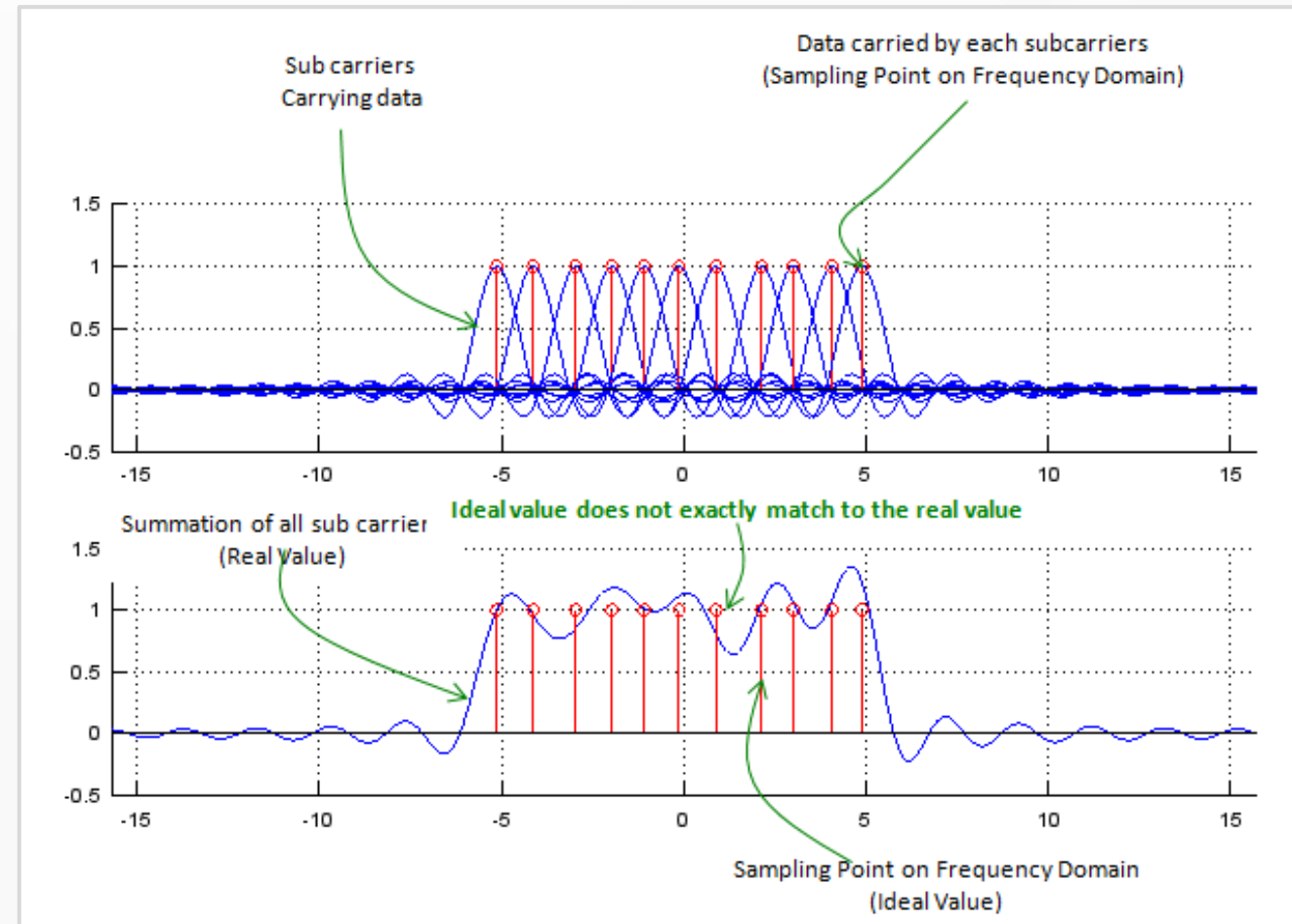
- ✓ Proper spacing between subcarriers is necessary to maintain orthogonality.
- ✓ if the space between sub carriers are not maintained accurately and they are drifting around, it can affect performance.
- ✓ When plotted separately, individual subcarriers may not show noticeable differences.
- ✓ When summed together, the impact of frequency drift becomes evident.



Orthogonal Frequency Division Multiplexing (OFDM)

OFDM

- ✓ In reality, frequency drift always exists due to environmental factors causes include fading and the Doppler effect.
- ✓ In reality there is no such an environment in which there is no frequency drift.
- ✓ In OFDM, you have to determine the frequency space in which the system can tolerate the signal distortion (تحمل تشويه الإشارة) due to frequency drift of sub carriers.



Orthogonal Frequency Division Multiplexing (OFDM)

- ☑ You can experiment with OFDM and frequency drift using the following Python code. Try different values for **No Of Carriers** and **fnoise Max** to observe their effects.

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
NoOfCarriers = 5 # Must be an odd number to ensure symmetric carrier distribution
f = np.arange(-5*np.pi, 5*np.pi, np.pi/50) # Frequency range from  $-5\pi$  to  $5\pi$  with small increments
fnoiseMax = 0.3 # Maximum frequency noise deviation

iMin = -(NoOfCarriers-1)//2 # Minimum carrier index

iMax = (NoOfCarriers-1)//2 # Maximum carrier index
csum = np.zeros(len(f)) # Initialize cumulative sum array with zeros

fList = [] # List to store frequency shifts
cList = [] # List to store maximum values of sinc functions
```

Orthogonal Frequency Division Multiplexing (OFDM)

- ☑ You can experiment with OFDM and frequency drift using the following Python code. Try different values for **No Of Carriers** and **fnoise Max** to observe their effects.

```
# Plot individual sinc functions
plt.figure(figsize=(10, 6)) # Set figure size
plt.subplot(2, 1, 1) # Create a subplot with 2 rows, 1 column, and select the first plot
plt.grid() # Enable grid for better visibility

for i in range(iMin, iMax+1): # Loop through carrier indices
    fnoise = fnoiseMax * (np.random.rand() - 0.5) # Generate small random frequency noise
    fshift = (i * (1/np.pi) * np.pi) + fnoise # Compute frequency shift with noise
    c = np.sinc(f - fshift) # Compute sinc function centered at the shifted frequency
    csum += c # Accumulate sinc functions
    fList.append(fshift) # Store frequency shift
    cList.append(max(c)) # Store maximum value of sinc function
    plt.plot(f, c) # Plot individual sinc function
    plt.stem([fshift], [1], 'r-', markerfmt='ro') # Mark peak positions with red stems
```

Orthogonal Frequency Division Multiplexing (OFDM)

- ☑ You can experiment with OFDM and frequency drift using the following Python code. Try different values for **No Of Carriers** and **fnoise Max** to observe their effects.

```
plt.axis([min(f), max(f), -0.5, 1.5]) # Set axis limits

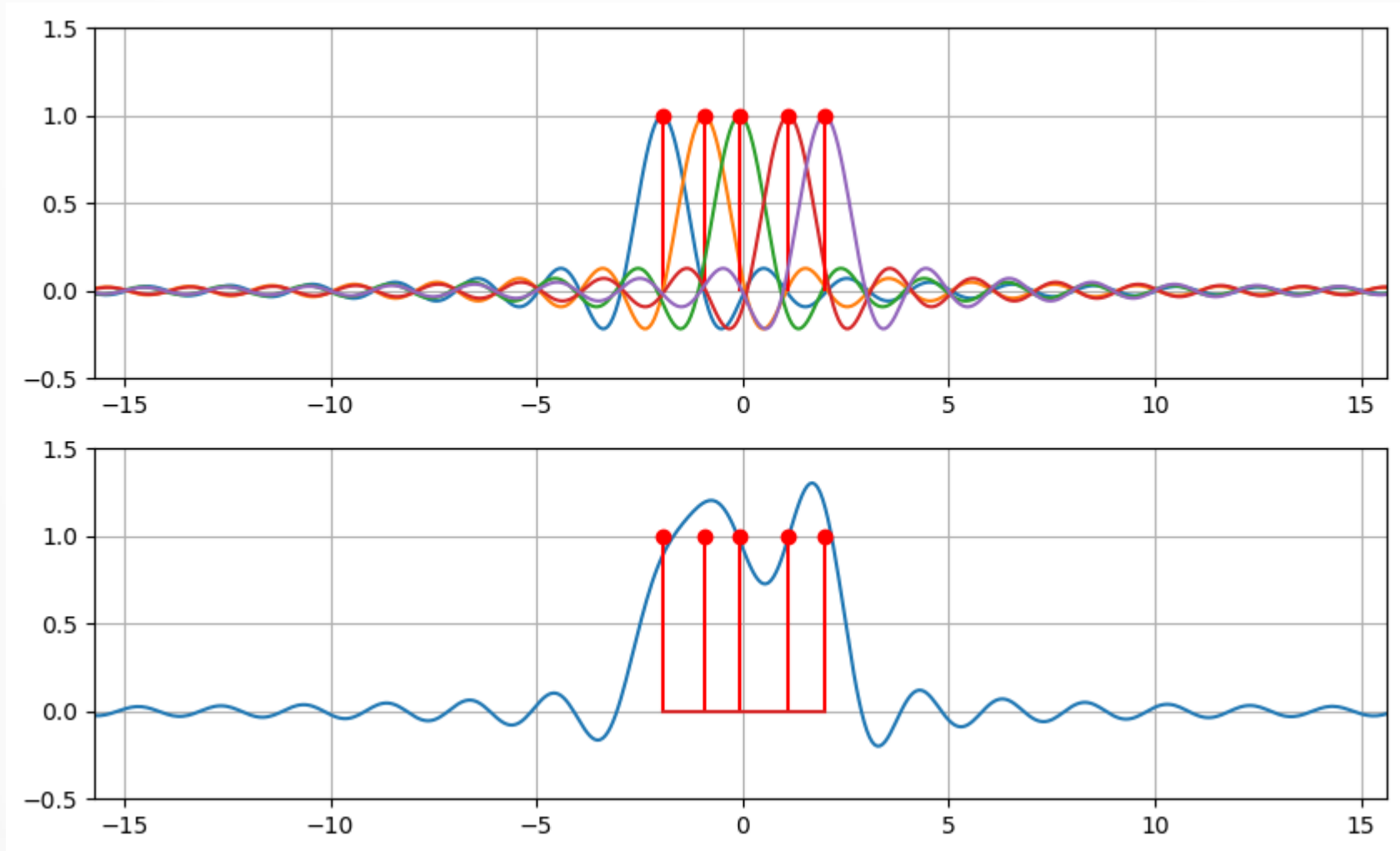
# Plot cumulative sum
plt.subplot(2, 1, 2) # Select second subplot
plt.grid() # Enable grid
plt.plot(f, csum) # Plot cumulative sum of sinc functions
plt.stem(fList, cList, 'r-', markerfmt='ro') # Mark peak positions with red stems
plt.axis([min(f), max(f), -0.5, 1.5]) # Set axis limits

plt.show() # Display the plot
```

Orthogonal Frequency Division Multiplexing (OFDM)

No of carriers = 3

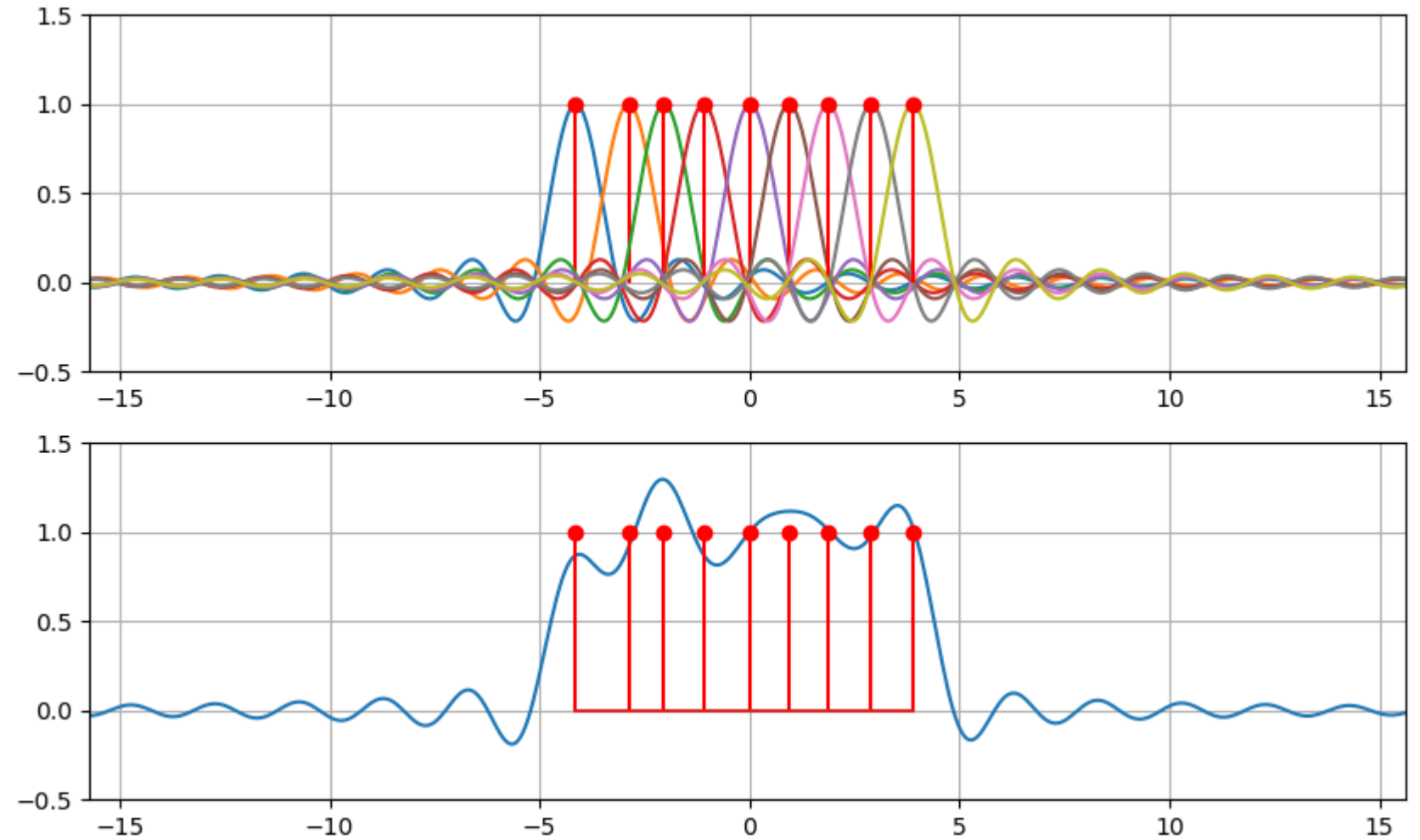
F noise Max = 0.3



Orthogonal Frequency Division Multiplexing (OFDM)

No of carriers = 9

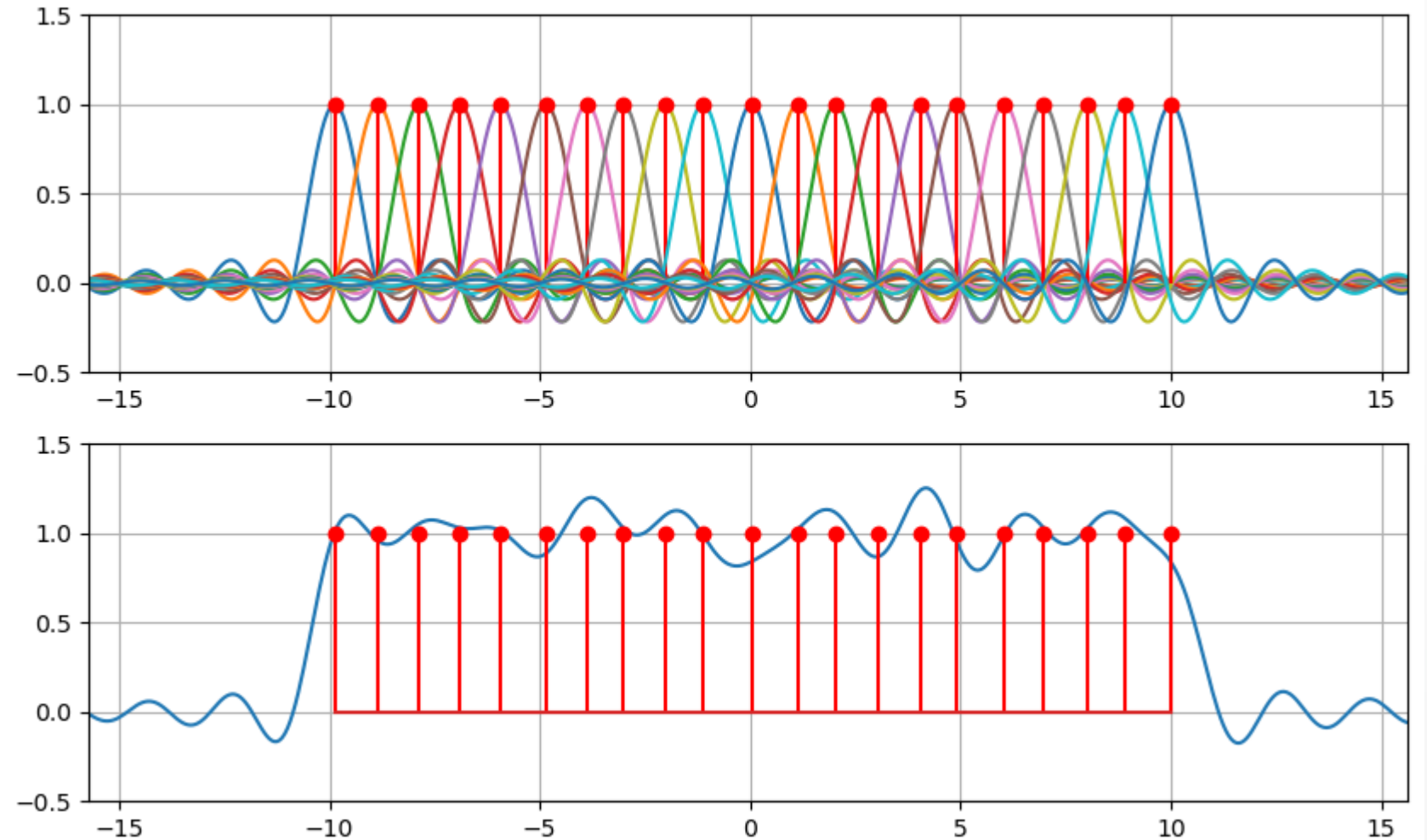
F noise Max = 0.3



Orthogonal Frequency Division Multiplexing (OFDM)

No of carriers = 21

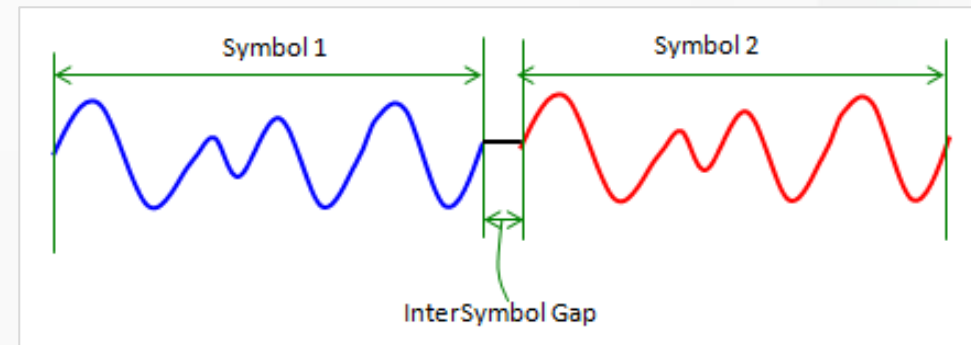
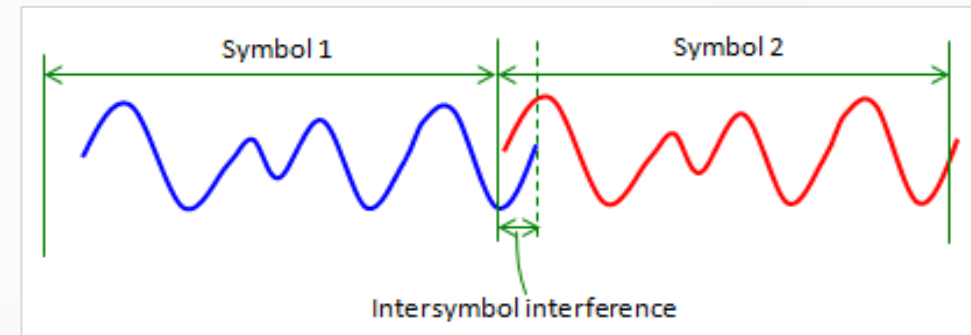
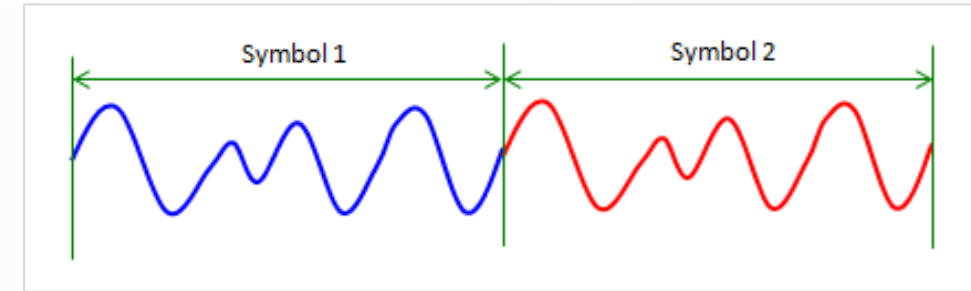
F noise Max = 0.3



Orthogonal Frequency Division Multiplexing (OFDM)

Cyclic Prefix

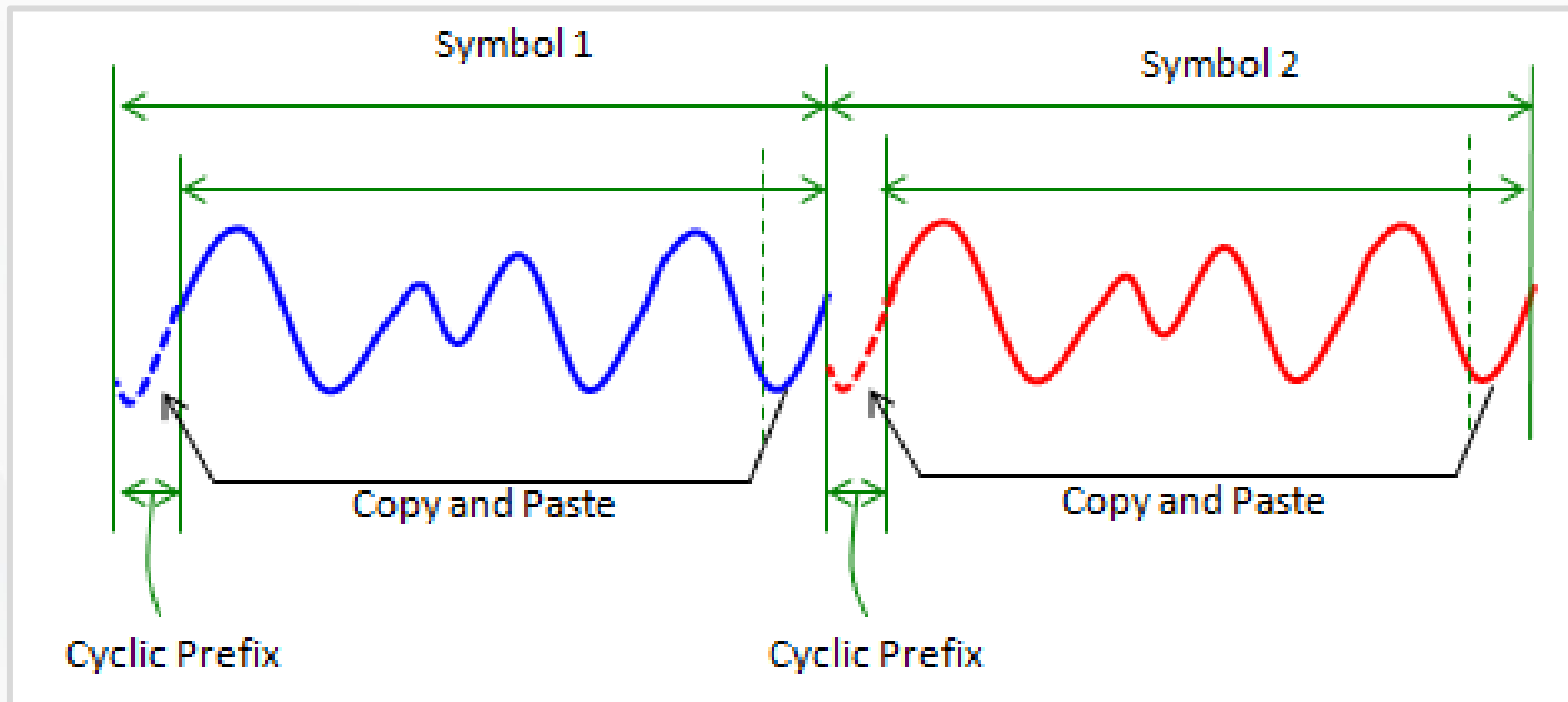
- ✓ The corresponding figure illustrates two OFDM symbols in sequence.
- ✓ In an ideal case, the signal has no issues. However, if the first symbol is delayed, its end overlaps with the next symbol, causing **Inter-Symbol Interference** (ISI).
- ✓ The solution is to design the system to handle delays since the radio channel cannot be controlled. A simple approach is adding a time **gap between** symbols to prevent Inter-Symbol Interference (ISI).



Orthogonal Frequency Division Multiplexing (OFDM)

Cyclic Prefix

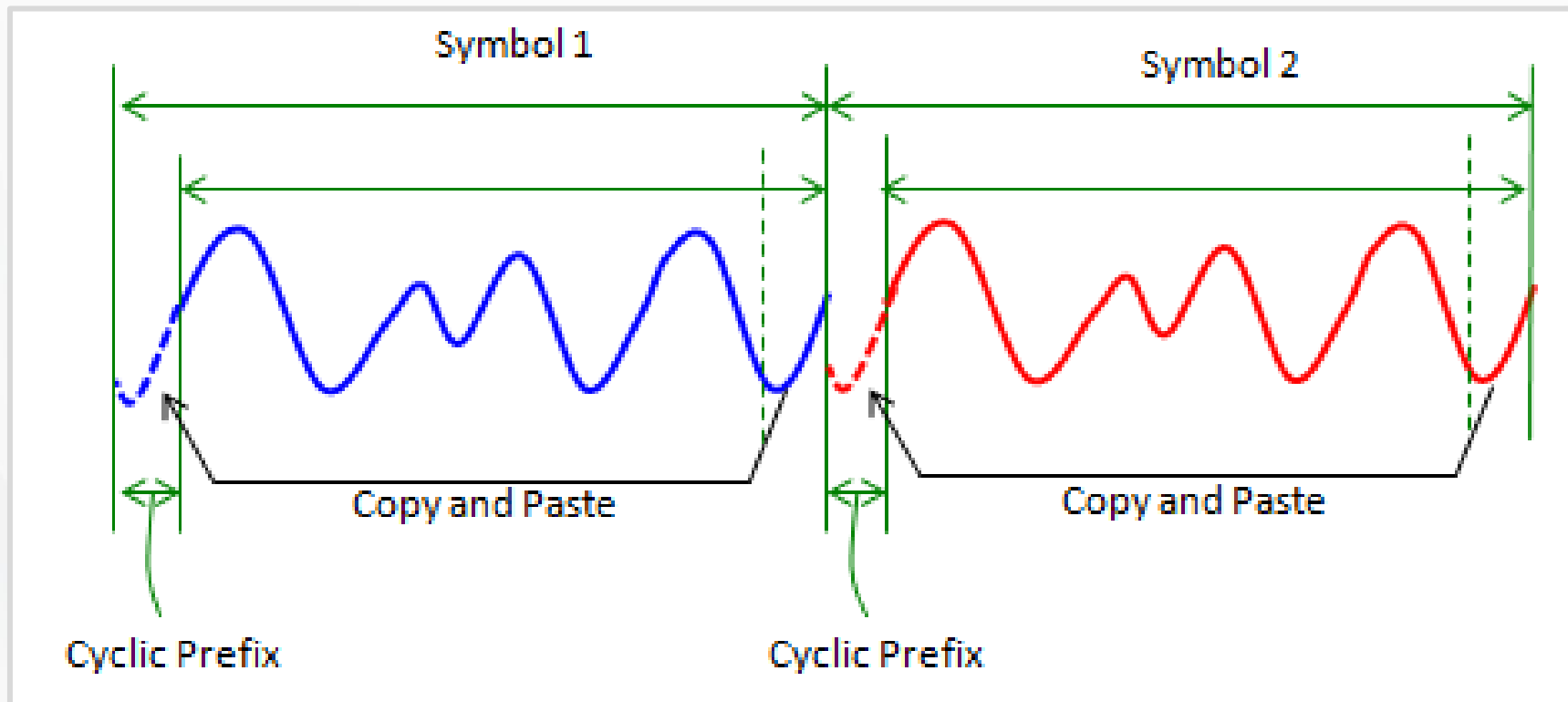
- ☑ To handle delay and intersymbol interference, a gap is added between symbols. However, leaving it empty can cause amplifier issues. To avoid this, a portion of the signal's end is copied and placed in the gap, forming a Cyclic Prefix.



Orthogonal Frequency Division Multiplexing (OFDM)

Cyclic Prefix

- ☑ To handle delay and intersymbol interference, a gap is added between symbols. However, leaving it empty can cause amplifier issues. To avoid this, a portion of the signal's end is copied and placed in the gap, forming a Cyclic Prefix.



THANKS

